

# Internship Report

On

Experiments towards comparing Document clustering methods

At



Centre for Artificial Intelligence & Robotics  
DRDO Complex, CV Raman Nagar  
Bengaluru-560093

*in partial fulfillment of the degree*

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in  
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Harcourt Butler Technical University  
Kanpur, Uttar Pradesh



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# About the Institution

## Defence Research and Development Organisation

The Defence Research and Development Organisation (DRDO) is an agency under the Department of Defence Research and Development in India, charged with the military's research and development. Established in 1958, DRDO has grown to be a network of over 50 laboratories, each engaged in developing defense technologies across various disciplines, including aeronautics, armaments, electronics, combat vehicles, engineering systems, instrumentation, missiles, advanced computing, simulation, and life sciences. The organization's mission is to achieve self-reliance in critical defense technologies and systems, and to equip India's armed forces with state-of-the-art weaponry and equipment. DRDO has been pivotal in developing indigenous systems like the Agni and Prithvi missiles, Tejas fighter aircraft, and the Arjun tank, significantly contributing to the nation's defense capabilities and technological prowess. Additionally, DRDO's innovations extend beyond traditional defense, encompassing works like the development of biogas technology to address sanitation issues and collaborations with the health-care sector to produce affordable medical devices. Through its extensive research and development efforts, DRDO plays a crucial role in enhancing India's strategic autonomy and bolstering national security.

## Centre for Artificial Intelligence and Robotics

The Centre for Artificial Intelligence and Robotics (CAIR) is a premier research institute in India under the Defence Research and Development Organisation (DRDO). Established to drive advancements in artificial intelligence (AI) and robotics, CAIR focuses on developing cutting-edge technologies that enhance defense capabilities and national security. The center is

involved in research and development works spanning intelligent systems, robotics, command and control systems, and communication networks. By leveraging AI, machine learning, and advanced robotics, CAIR aims to create autonomous systems that can perform complex tasks in dynamic environments, thereby reducing human intervention and increasing operational efficiency. Through its innovative work, CAIR plays a vital role in modernizing India's defense infrastructure, contributing to the nation's technological self-reliance, and addressing emerging security challenges.

# Acknowledgement

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# Abstract

This report presents a detailed analysis of our internship experience at CAIR, DRDO. The primary objective of the internship was to develop a workflow for optimizing document clustering using various word embedding techniques. This workflow aims to overcome the challenges associated with document clustering by employing different word embedding methods. In this work, we tackle the issue of refining document clustering by employing multiple word embedding techniques. Traditional document analysis methods, which depend heavily on manual review, are both time-consuming and inefficient, especially when handling large datasets with numerous duplicates and variations. Our approach utilizes Natural Language Processing (NLP) to automate document classification and clustering. The work assesses and contrasts several word embedding techniques, such as Bag of Words, TF-IDF, Word2Vec, and GloVe. We developed a co-occurrence matrix and applied TF-IDF vectorization combined with Principal Component Analysis (PCA) and K-Means clustering. This method facilitates the identification of high co-occurrence word pairs and singleton words, enabling more precise clustering and visualization of document similarities. The ultimate goal is to establish a robust system for document analysis that can accurately group similar documents. This research contributes to the field of NLP by providing insights into the effectiveness of different word embedding techniques and their application in document clustering, paving the way for more advanced and automated document analysis solutions.

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# Chapter 1

## Introduction

In the digital world, the analysis of large volumes of text data is crucial. The objective of this study is to develop a robust system for document clustering and analysis that integrates feature extraction, pre-processing, word embedding, clustering algorithms, and visualization. By leveraging advanced techniques such as Word2Vec, GloVe, and clustering methods, the system is designed to enhance the accuracy and efficiency of text data analysis. Additionally, the implementation of co-occurrence matrices provides deeper insights into word relationships within the text.

### 1.1 Background Information

In today's digital world, analyzing large volumes of textual data is crucial. Traditional methods of text analysis often fall short when dealing with the complexity and scale of contemporary datasets, making advanced natural language processing (NLP) techniques necessary. Data Extraction involves pulling text from different document formats to ensure accurate data for analysis. Data Preprocessing, which includes steps like tokenization, normalization, stopwords removal, and stemming, cleans and standardizes the text. Techniques such as Bag of Words (BoW), TF-IDF, Word2Vec, and GloVe in Word Embedding convert text into meaningful forms, capturing semantic and contextual nuances. Clustering Algorithms, especially K-Means, categorize similar documents to uncover underlying themes and patterns.

This study utilizes modern web frameworks for visualization, offering an efficient and intuitive way to present meaningful insights derived from various documents. The integration of these frameworks ensures that the visual representation of clustering results is both interactive and user-friendly, facilitating better understanding and interpretation of the data. Our approach aims to streamline the process of document analysis, making it more accessible and effective for handling large and complex datasets.

## 1.2 Literature Survey

In our study, we deeply discussed multiple features of word embedding techniques and clustering algorithms in Natural Language Processing (NLP), utilizing well-recognized sources to guide our research and implementation. The literature provides a comprehensive background on these topics, highlighting their development, applications, and evaluation metrics.

**Word Embedding Techniques:** Word embedding techniques have revolutionized the field of NLP by enabling the representation of words in continuous vector spaces. These techniques capture the semantic relationships between words, allowing for more meaningful analysis and processing of textual data. A particularly insightful resource was the blog post "Guide on Word Embeddings in NLP" from Turing, which provided a thorough introduction to word embeddings and their significance. This post discussed various embedding techniques such as Word2Vec and GloVe, detailing their advantages and applications in NLP tasks. The literature underscores the importance of evaluating clustering performance using metrics like the Silhouette Score and Davies-Bouldin Index. These metrics provide insights into the compactness and separation of clusters, aiding in the selection of the most appropriate algorithms. Our study benefited from this body of knowledge, as we utilized these metrics to assess the effectiveness of our clustering methods.

**Co-occurrence Matrix:** In addition to word embeddings and clustering algorithms, we incorporated the use of a co-occurrence matrix to enhance our analysis. A co-occurrence matrix captures the frequency with which pairs of words appear together in a corpus, providing valuable insights into the relationships between words. This technique has been shown to improve the accuracy of clustering by identifying frequently co-occurring terms and their contextual significance.

The integration of the co-occurrence matrix with word embedding techniques and clustering algorithms is supported by numerous studies. These studies highlight how co-occurrence matrices can uncover hidden relationships within the text, complementing the semantic insights provided by word embeddings. This combined approach has been effective in producing more accurate and interpretable clusters, as documented in various research papers and articles.

### NLP Pipeline

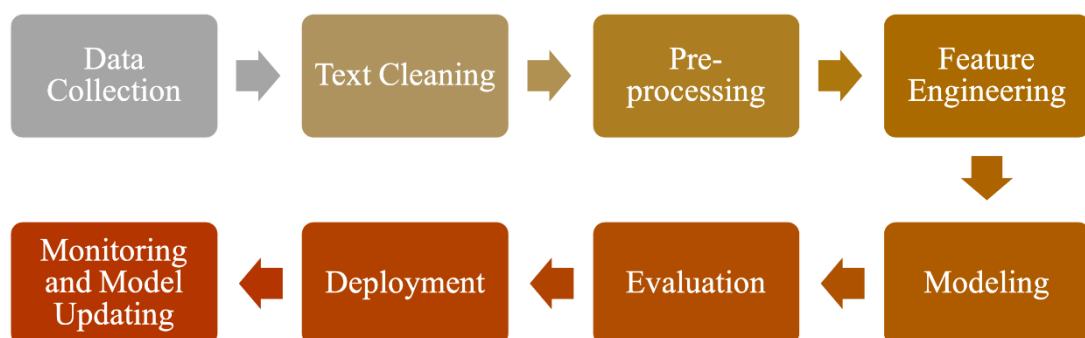


Figure 1.1: NLP Pipeline



### Association Rule Mining (ARM):

Association rule mining (ARM) is a technique used to discover interesting relationships, patterns, or associations among a set of items in large datasets. It is often used in market basket analysis, where the goal is to find associations between products that frequently co-occur in transactions.

#### Algorithm Names:

Apriori Algorithm : : It generates candidate itemsets and uses a priori knowledge to reduce the number of itemsets to be evaluated.

FP-Growth Algorithm: It constructs a data structure called FP-tree to efficiently mine frequent itemsets without generating candidate itemsets explicitly.

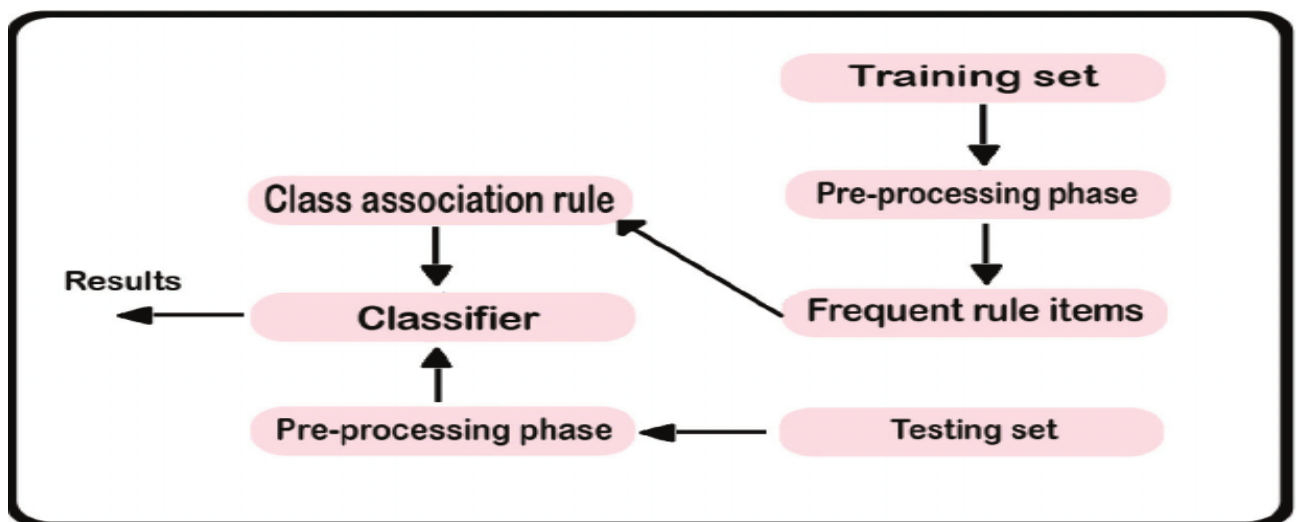


Figure1.2 : Flow chart of association rule generation

## 1.3 Problem Statement

Analyzing documents with multiple copies and variations by reading the content is a tedious task. Checking for novelty, changes in the content of documents, or similarity between documents can be done if there is a method or system that could group the documents. This work implements a user-driven approach that could be used to clustering documents effectively.

This work aims to evaluate and compare the effectiveness of various word embedding techniques, including Bag of Words, TF-IDF, Word2Vec, and GloVe, as well as the application co-occurrence matrices in clustering and analyzing textual data. By leveraging performance metrics such as the Silhouette Score and Davies-Bouldin Index, the work seeks to identify the most suitable methods for accurately capturing semantic relationships, improving text clustering, and extracting meaningful patterns in natural language processing applications.

Additionally, this work includes the development of a co-occurrence matrix and the application of TF-IDF vectorization combined with PCA and KMeans clustering to enhance document analysis. This method helps in identifying high co-occurrence word pairs and singleton words, facilitating the clustering and visualization of document similarities.

# Chapter 2

## Methodology

Our work focuses on enhancing document clustering by evaluating and comparing various word embedding techniques, including Bag of Words, TF-IDF, Word2Vec, and GloVe, to improve the accuracy and efficiency of document analysis. By incorporating association rule mining (ARM) and co-occurrence matrices, we aim to capture semantic relationships within the text and optimize clustering performance. The work utilizes performance metrics like the Silhouette Score and Davies-Bouldin Index to determine the most effective techniques, ultimately aiming to streamline document-intensive workflows by accurately grouping similar documents and highlighting novel changes.

### 2.1 Data Processing Workflow

#### 2.1.1 Data Collection and Preprocessing:

Gathering a diverse set of documents for analysis.

Perform text preprocessing, including tokenization, stopwords removal, stemming, and lemmatization, using NLTK

#### 2.1.2 Word Embedding Techniques:

##### a) Bag of words

The bag of words model is a simple document embedding technique based on word frequency. Conceptually, we think of the whole document as a “bag” of words, rather than a sequence. We represent the document simply by the frequency of each word. For example, if we have a vocabulary of 1000 words, then the whole document will be represented by a 1000-dimensional vector, where the vector’s  $i$ th entry represents the frequency of the  $i$ th vocabulary word in the document.

Using this technique, we can embed a whole set of documents and feed them into a variety of different machine-learning algorithms. But it does for simple classification problems, and its simplicity and ease of use make it an attractive choice.

## **b) Bag of n grams**

A bag-of-n-grams is a technique used in natural language processing (NLP) to represent a piece of text. It's an extension of the bag-of-words model, which treats a text as a collection of individual words, ignoring their order.

### **Different types of n-grams:**

**Unigrams (1-grams):** Unigrams are single words considered individually. For example, in the sentence "I love natural language processing," the unigrams would be ['I', 'love', 'natural', 'language', 'processing'].

**Bigrams (2-grams):** Bigrams consist of sequences of two adjacent words. For the same sentence, the bigrams would be ['I love', 'love natural', 'natural language', 'language processing'].

**Trigrams (3-grams):** Trigrams are sequences of three adjacent words. Continuing with the example sentence, the trigrams would be ['I love natural', 'love natural language', 'natural language processing'].

**N-grams (N-grams):** N-grams refer to sequences of N adjacent elements, which can be words, characters, or symbols. For instance, if 'N' is 4, then we have 4-grams, also known as quadgrams

## **c) Word2 Vec**

Word2Vec is a widely used method in natural language processing (NLP) that allows words to be represented as vectors in a continuous vector space. Word2Vec is an effort to map words to high-dimensional vectors to capture the semantic relationships between words, developed by researchers at Google. Words with similar meanings should have similar vector representations, according to the main principle of Word2Vec.

## **d) Glove**

GloVe, which stands for Global Vectors for Word Representation, is a word embedding technique. It aims to capture the meaning of words in a vector space such that words with similar meanings are positioned close to each other. Unlike other embedding methods that use local context (like Word2Vec), GloVe combines both local context and global statistical information about a corpus.

## e) TF-IDF (Term Frequency-Inverse Document Frequency)

It is a numerical statistic that is used to reflect how important a word is to a document in a collection or corpus. It's commonly used in information retrieval and text mining.

Term Frequency (TF) Measures how frequently a term appears in a document. It is calculated as the ratio of the number of times a term appears in a document compared to the total number of terms in that document. Inverse Document Frequency (IDF) Measures how important a term is across the entire corpus. It is calculated as the logarithm of the ratio of the total number of documents to the number of documents containing the term, adjusted for terms that appear in all documents (to avoid division by zero).

### 2.1.3 Co-Occurrence Matrix Construction:

We have created co-occurrence matrix with **two logical definitions**. The first one is as follows:

- **First Method**

A co-occurrence matrix is a square matrix used to represent the co-occurrence (i.e., the frequency with which two words appear near each other) of words in a corpus of documents.

Given a set of documents and a specified window size, the co-occurrence matrix captures how many times each pair of words appears within the specified window in any document.

#### Construction of Co-Occurrence Matrix

1. Vocabulary Extraction:
  - Extract a list of unique words (vocabulary) from the preprocessed text documents.
2. Initialize the Matrix:
  - Create a square matrix of size vocab\_size x vocab\_size (where vocab\_size is the number of unique words in the vocabulary).
  - Each cell (i, j) in the matrix will eventually hold the co-occurrence count of the words represented by i and j indices in the vocabulary.
3. Populate the Matrix:
  - Slide a window of specified size over each document.
  - For each word within the window, increment the count for every pair of words within that window in the matrix.

#### Explanation of the Matrix

- The rows and columns of the matrix represent words from the vocabulary.
- Each cell (i, j) represents the number of times the word vocab[i] appears within the window size of the word vocab[j].

- **Second Method**

#### **Co-Occurrence Matrix Definition with Distance Weighting**

In this revised version of the co-occurrence matrix function, each co-occurrence is weighted by the inverse of the distance between the words. This means that closer words within the specified window size contribute more to the co-occurrence count than words further apart.

#### **Formal Definition**

Given a corpus of text documents and a specified window size  $W$ , the co-occurrence matrix

$M$  is defined as follows:

Let  $V$  be the vocabulary of unique words in the corpus.

Let  $n$  be the size of the vocabulary ( $|V|$ ).

The co-occurrence matrix  $M$  is an  $n \times n$  matrix where each element  $M[i, j]$  represents the weighted count of how often the word  $V[i]$  appears within a window of  $W$  words before or after the word  $V[j]$  in any of the documents in the corpus. The weight is calculated as  $1.0 / \text{distance}$ , where distance is the number of words between  $V[i]$  and  $V[j]$ .

### **2.1.4 Dimensionality Reduction:**

#### **Principal Component Analysis (PCA)**

Principal Component Analysis (PCA) is a statistical technique used to simplify the complexity in high-dimensional data while retaining trends and patterns. It achieves this by transforming the data into a new set of variables called principal components. These components are ordered such that the first few retain most of the variation present in the original dataset.

### **2.1.5 K-Means Clustering :**

K-Means is a widely used clustering algorithm that partitions data into  $k$  distinct clusters. The goal is to minimize the variance within each cluster.

The k-means clustering algorithm begins with the initialization step, where  $k$  cluster centroids are chosen randomly. During the assignment step, each data point is assigned to the nearest centroid, thus forming  $k$  clusters. The centroids are then updated by recalculating them as the mean of all points in each cluster. These assignment and update steps are repeated iteratively until the centroids stabilize or a predefined number of iterations is reached, ensuring that the clusters are as compact and distinct as possible.

#### **To determine the optimal number of clusters we use Elbow Method**

The Elbow Method is a widely used technique to determine the optimal number of clusters in K-Means clustering. This method involves plotting the within-cluster sum of squares (WCSS) against the number of clusters ( $k$ ). WCSS measures the total variance within each cluster, indicating how closely the data points are grouped. As the number of clusters increases, WCSS decreases because the clusters become smaller and more compact. To apply the Elbow Method, one runs the K-Means algorithm for a range of  $k$  values, typically from 1 to a reasonable maximum. For each  $k$ , the WCSS is calculated and plotted, with the number of clusters on the x-axis and WCSS on the y-axis.

Initially, as more clusters are added, WCSS decreases significantly since data points are grouped more tightly. However, after a certain point, the rate of decrease slows, creating a distinct bend or "elbow" in the plot. This "elbow" point suggests the optimal number of clusters, balancing the trade-off between having too few clusters, which results in large, heterogeneous clusters with high WCSS, and having too many clusters, which can lead to overfitting and unnecessary complexity. By selecting the number of clusters at the elbow, one achieves a good compromise between minimizing WCSS and maintaining a simple model. However, it's worth noting that the Elbow Method is heuristic and may not always present a clear elbow, requiring additional methods or domain expertise to determine the best number of clusters.

#### **Evaluation Metrics**

Evaluation metrics are quantitative measures used to assess the performance and quality of machine learning models or clustering algorithms. These metrics help in determining how well a model or algorithm captures the underlying patterns in the data and meets the desired objectives.

### ***Silhouette Score***

The Silhouette Score measures the quality of clustering by comparing the distance between data points and their own cluster versus other clusters. It ranges from -1 to 1

- **1:** Well-clustered.
- **0:** On the boundary.
- **-1:** Mis-clustered.

### ***Davies-Bouldin Index***

The Davies-Bouldin Index (DBI) evaluates clustering quality by comparing the ratio of within-cluster dispersion to between-cluster separation. Lower DBI values indicate better clustering.

- **Low DBI (close to 0):** Good clustering with compact and well-separated clusters.
- **High DBI:** Poor clustering with overlapping or spread-out clusters.



# Chapter 3

## Implementation

### 3.1 Programming Language and Tools

- Justification for Using Python:

Python was chosen for this work due to its extensive ecosystem of libraries and frameworks that facilitate natural language processing (NLP) and data analysis. Its simplicity and readability make it an ideal choice for rapid development and prototyping. Additionally, Python's robust community support ensures access to a wealth of resources and tools for implementing various NLP techniques and clustering algorithms.

- Overview of the Libraries and Tools Used:

The work utilized several key Python libraries:

1. Numpy and Pandas: For data manipulation and handling.
2. Scikit-learn: For implementing clustering algorithms (K-means) and preprocessing techniques (Bag of Words, TF-IDF).
3. Gensim: For Word2Vec embeddings.
4. Spacy: For advanced NLP tasks and word embeddings like GloVe.
5. Matplotlib : For data visualization to evaluate clustering results. These libraries provided a comprehensive set of tools to preprocess, analyze, and visualize text data effectively.

## **3.2 About the Dataset**

The dataset used in this project consists of five documents covering different subjects: Natural Language Processing (NLP), Artificial Intelligence (AI), Machine Learning (ML), Diet, and Sleep Schedule. Each document provides insights into its respective topic, ranging from technological advancements and applications to health and wellness.

# Chapter 4

## Experimental Results

### 4.1 Text Preprocessing

```
import re # Import regular expression module
import string # Import string manipulation module
from bs4 import BeautifulSoup # Import BeautifulSoup for HTML parsing
from nltk.corpus import stopwords # Import NLTK's stopwords corpus
from nltk.tokenize import word_tokenize # Import NLTK's word tokenizer
import nltk # Import NLTK for natural language processing utilities

# Download NLTK resources (if not already downloaded)
nltk.download('stopwords')
nltk.download('punkt')

1 usage
def preprocess_text(text):
    # Remove HTML tags using BeautifulSoup
    text = BeautifulSoup(text, features="html.parser").get_text()
    text = text.lower() # Convert text to lowercase
    text = re.sub(pattern=r'http\S+', repl='', text) # Remove URLs
    text = re.sub(pattern=r'\w*\d\w*', repl='', text) # Remove alphanumeric words
    text = re.sub(pattern=r'\d+', repl='', text) # Remove standalone numbers
    text = text.translate(str.maketrans('', '', string.punctuation)) # Remove punctuation
    stop_words = set(stopwords.words('english')) # Set of English stopwords
    tokens = word_tokenize(text) # Tokenize the text into words
    tokens = [word for word in tokens if word not in stop_words] # Remove stopwords
    preprocessed_text = ' '.join(tokens) # Join remaining tokens back into a string
    return preprocessed_text

# Sample HTML text for preprocessing
sample_text = """
<html>
  <body>
    <p>This is a <b>sample</b> text with HTML tags and a URL: https://example.com.</p>
    <p>It will be used for text preprocessing, including <i>removal</i> of stopwords, punctuation, and numbers like 123!</p>
  </body>
</html>
"""

# Process the sample text
processed_text = preprocess_text(sample_text)
print("Processed Text:", processed_text)
```

## Output

```
"C:\Program Files\Python312\python.exe" "C:\Users\ritik\PycharmProjects\pythonProject4\co_chotu.py"
[nltk_data] Downloading package stopwords to
[nltk_data]   C:\Users\ritik\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\ritik\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
Processed Text: sample text html tags url used text preprocessing including removal stopwords punctuation numbers like

Process finished with exit code 0
|
```

## 4.2 Co-occurrence matrix

```
Processing Document 'Doc1_NLP.docx'...

Co-occurrence Matrix:
      analyzing  patterns  goal  ai  generate
analyzing      0         0     0   0         0
patterns      0         0     0   0         0
goal          0         0     0   0         0
ai            0         0     0   0         0
generate      0         0     0   0         0

Processing Document 'Doc2_AI.docx'...

Co-occurrence Matrix:
      recent  advancements  leveraging  performance  ai
recent      0             1           0           0     0
advancements 1             0           0           0     0
leveraging   0             0           0           0     0
performance  0             0           0           0     0
ai           0             0           0           0     0

Processing Document 'Doc3_ML.docx'...

Co-occurrence Matrix:
      revolutionizing  interact  use  machine  ai
revolutionizing      0         0   0         0     0
interact             0         0   1         0     0
use                 0         1   0         0     0
```

```

0.00 0 1 0 0 0
machine 0 0 0 0 0
ai 0 0 0 0 0

Processing Document 'Doc4_Diet.docx'...

Co-occurrence Matrix:
      food  water  consuming  lean  essential
food      0      0      0      0      0
water      0      0      0      0      0
consuming  0      0      0      0      0
lean       0      0      0      0      0
essential  0      0      0      0      0

Processing Document 'Doc5_SleepSchedule.docx'...

Co-occurrence Matrix:
      body  typically  issues  time  poor
body      0      0      0      0      0
typically  0      0      0      0      0
issues     0      0      0      0      0
time       0      0      0      0      0
poor       0      0      0      0      0

Process finished with exit code 0

```

## Method 2

```

Co-occurrence Matrix (First few rows and columns):
      artificial  extract  interpret  computers  human
artificial      0.0      0.0      0.0      0.0      0.0
extract          0.0      0.0      0.0      0.0      0.0
interpret        0.0      0.0      0.0      0.5      0.5
computers        0.0      0.0      0.5      0.0      0.0
human            0.0      0.0      0.5      0.0      0.0

Processing Document 2...
4.3 Different Word Embedding Techniques

Co-occurrence Matrix (First few rows and columns):
      respond  artificial  comprehend  significant  human
respond      0.0      0.0      1.0      0.0      1.0
artificial    0.0      0.0      0.0      0.0      0.0
comprehend    1.0      0.0      0.0      0.0      0.5
significant    0.0      0.0      0.0      0.0      0.0
human         1.0      0.0      0.5      0.0      0.0

```

Processing Document 1...

Co-occurrence Matrix:

|         | large | goal | extract | amounts | enable |
|---------|-------|------|---------|---------|--------|
| large   | 0     | 0    | 0       | 1       | 0      |
| goal    | 0     | 0    | 0       | 0       | 1      |
| extract | 0     | 0    | 0       | 0       | 0      |
| amounts | 1     | 0    | 0       | 0       | 0      |
| enable  | 0     | 1    | 0       | 0       | 0      |

Pairs of Words with the Highest Co-occurrence Counts:

language - nlp: 3

language - human: 2

Singleton Words (based on co-occurrence):

large

goal

extract

amounts

enable

insights

humans

generate

artificial

...

tissues

quality

Most occurring word in Bag of Words PCA: adults

All words for Document 5: ['irregular', 'night', 'habits', 'sleep', 'adults']

## 4.3 Different Word Embedding Techniques

### 1. Bag of words :

Processing Document 1...

Co-occurrence Matrix:

|         | large | goal | extract | amounts | enable |
|---------|-------|------|---------|---------|--------|
| large   | 0     | 0    | 0       | 1       | 0      |
| goal    | 0     | 0    | 0       | 0       | 1      |
| extract | 0     | 0    | 0       | 0       | 0      |
| amounts | 1     | 0    | 0       | 0       | 0      |
| enable  | 0     | 1    | 0       | 0       | 0      |

Pairs of Words with the Highest Co-occurrence Counts:

language - nlp: 3

language - human: 2

Singleton Words (based on co-occurrence):

large  
goal  
extract  
amounts  
enable  
insights  
humans  
generate  
artificial  
...  
tissues  
quality

Most occurring word in Bag of Words PCA: adults

All words for Document 5: ['irregular', 'night', 'habits', 'sleep', 'adults']

Figure 4.1: Output

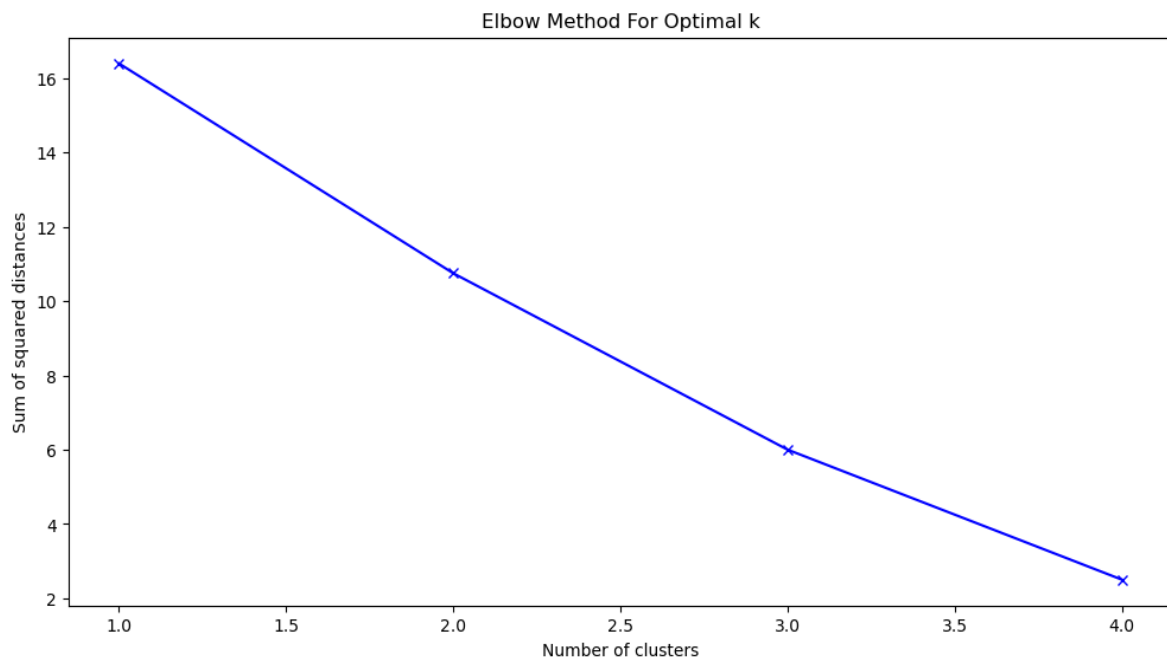


Figure 4.2 : Elbow Method for Optimal Number of Clusters

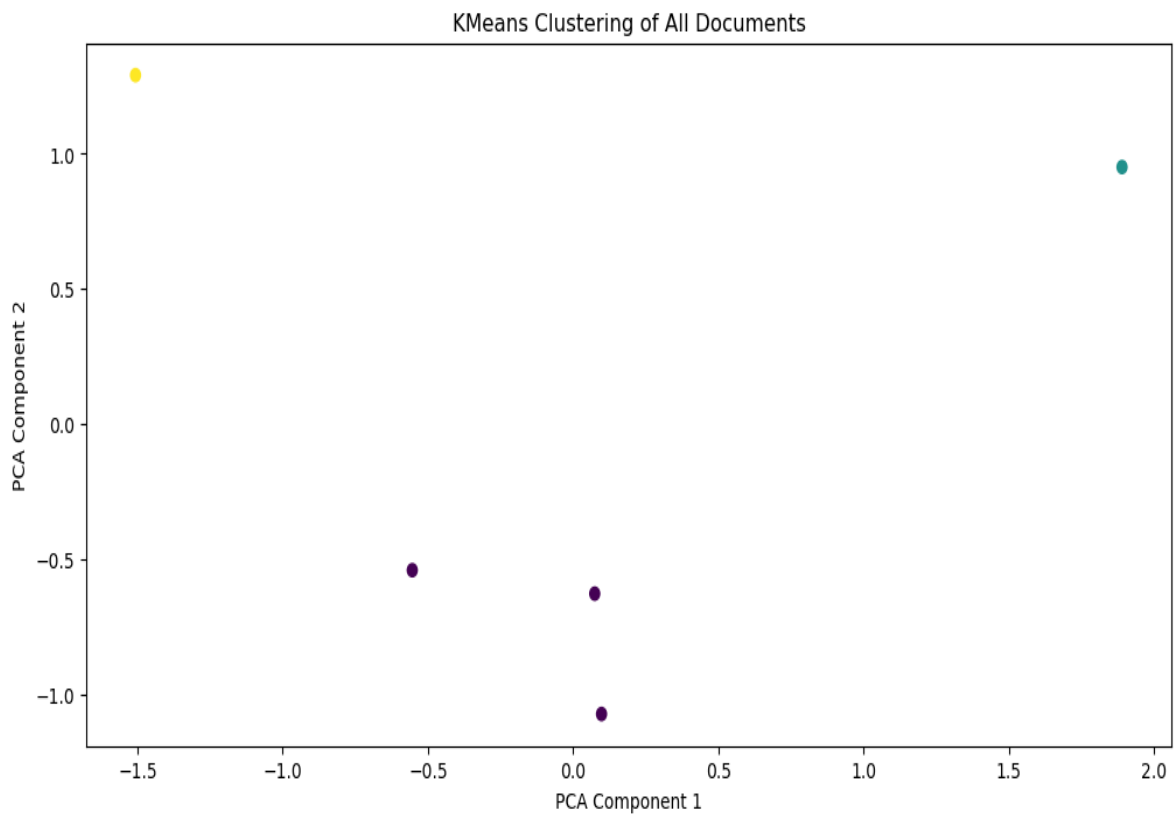


Figure 4.3: Plot of Documents



```

Document 'Doc1_NLP' is in Cluster 0
Document 'Doc2_AI' is in Cluster 1
Document 'Doc3_ML' is in Cluster 0
Document 'Doc4_Diet' is in Cluster 0
Document 'Doc5_SleepSchedule' is in Cluster 2
Silhouette Score: 0.09146923345074161
Davies-Bouldin Index: 0.5515175636450773

```

Figure 4.4: K-Means Clustering Results with Silhouette Score and Davies-Bouldin Index

## 2. TfidfVectorizer

```

Processing Document 1...
Co-occurrence Matrix:
      large goal extract amounts enable
large      0      0      0      1      0
goal      0      0      0      0      1
extract    0      0      0      0      0
amounts    1      0      0      0      0
enable     0      1      0      0      0

Pairs of Words with the Highest Co-occurrence Counts:
language - nlp: 3
language - human: 2

Singleton Words (based on co-occurrence):
large
goal
extract
amounts
enable
insights
humans
generate
artificial
...
tissues
quality
Most occurring word in TF-IDF PCA: adults
All words for Document 5: ['irregular', 'night', 'habits', 'sleep', 'adults']

```

Figure 4.5 : Output

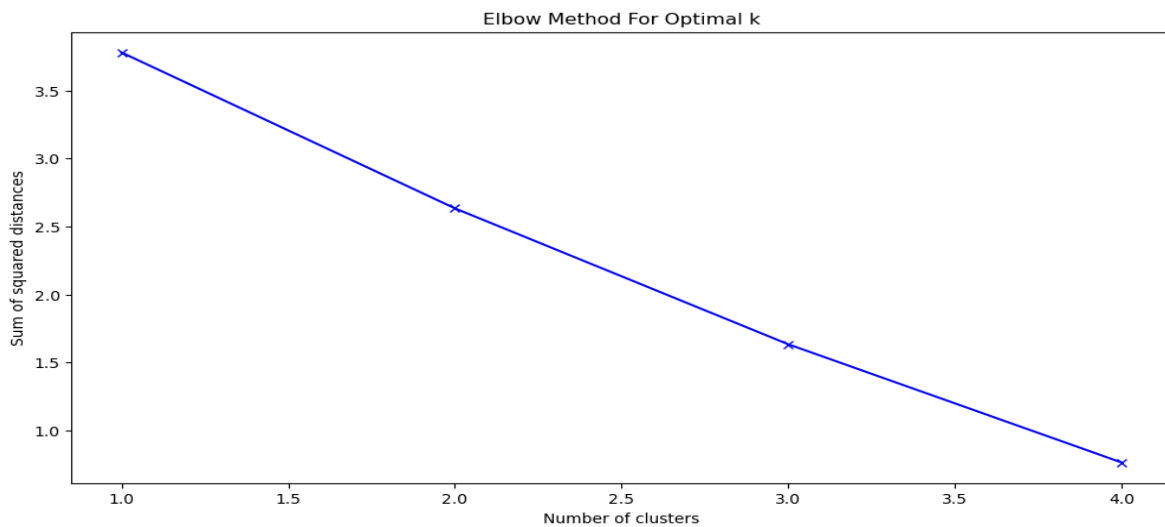


Figure 4.6 : Elbow Method for Optimal Number of Clusters

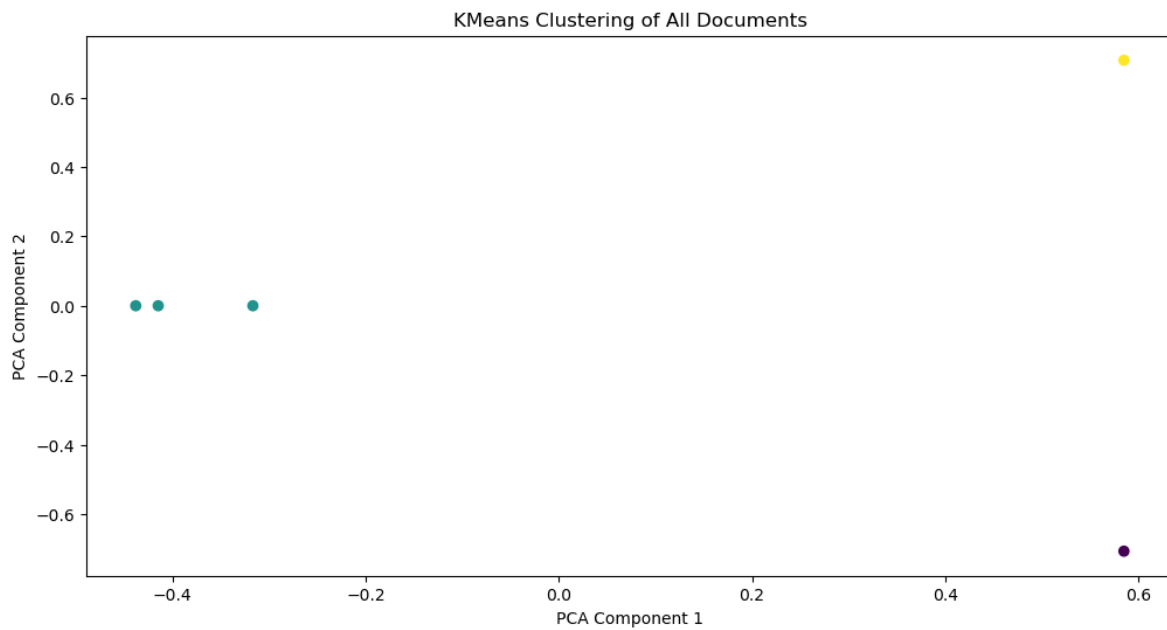


Figure 4.7: Plot of Documents

```
Document 'Doc1_NLP' is in Cluster 1
Document 'Doc2_AI' is in Cluster 1
Document 'Doc3_ML' is in Cluster 1
Document 'Doc4_Diet' is in Cluster 0
Document 'Doc5_SleepSchedule' is in Cluster 2
Silhouette Score: 0.05790751609142728
Davies-Bouldin Index: 0.6114689051314528
```

Figure 4.8: K-Means Clustering Results with Silhouette Score and Davies Index

### 3.Word2Vec

```
Processing Document 1...

Co-occurrence Matrix:
      enable  recognition  applications  nlp  interpret
enable      0            0            0    1          0
recognition  0            0            0    0          0
applications 0            0            0    1          0
nlp          1            0            1    0          0
interpret    0            0            0    0          0

Pairs of Words with the Highest Co-occurrence Counts:
nlp - language: 3
language - human: 2

Singleton Words (based on co-occurrence):
enable
recognition
applications
interpret
subfield
extract
amounts
ai
focuses
...
repair
chronic
Most occurring word in Word2Vec PCA: quality
All words for Document 5: ['sleep', 'night', 'irregular', 'habits', 'quality']
```

Figure 4.9 : Output

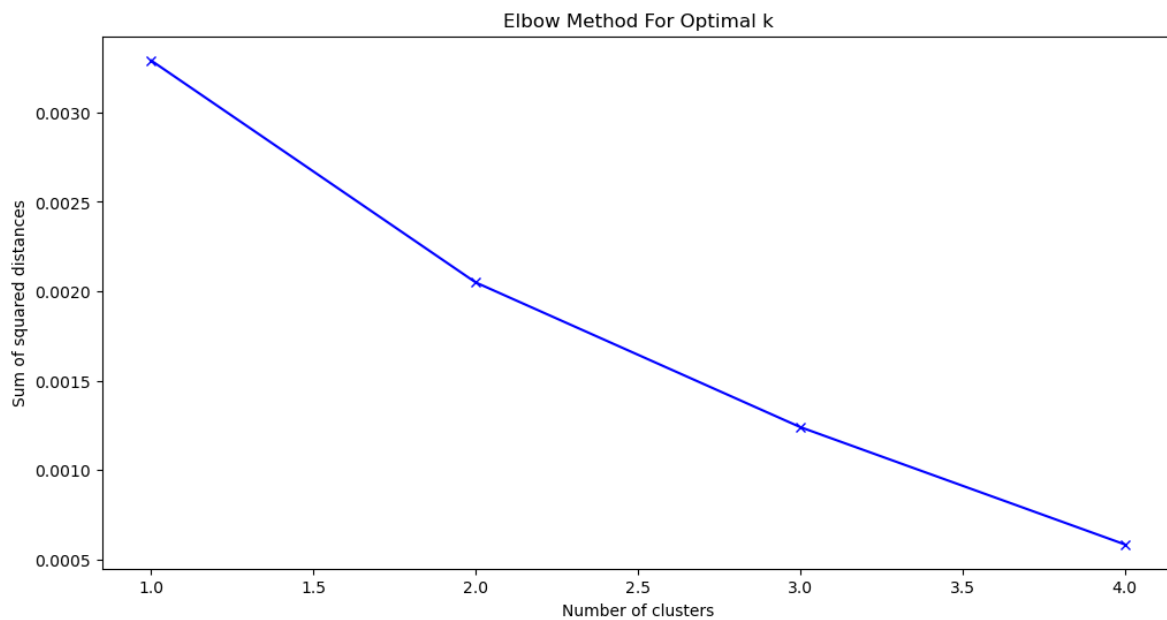


Figure 4.10 : Elbow Method for Optimal Number of Clusters

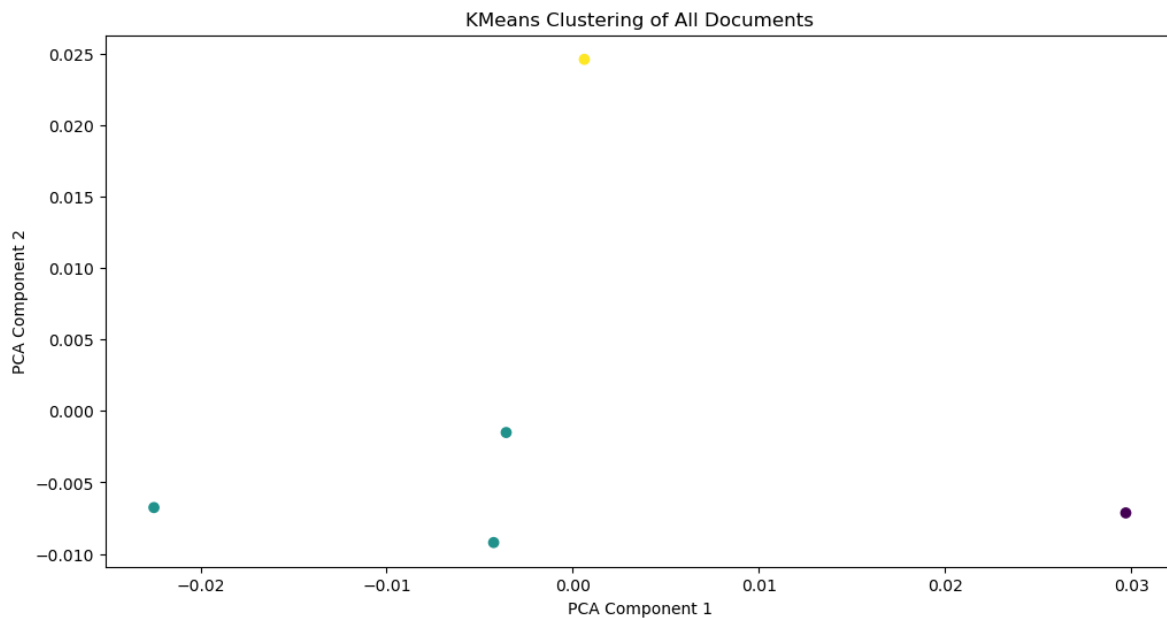


Figure 4.11 : Plot of Documents

```
Document 'Doc1_NLP' is in Cluster 1
Document 'Doc2_AI' is in Cluster 1
Document 'Doc3_ML' is in Cluster 1
Document 'Doc4_Diet' is in Cluster 0
Document 'Doc5_SleepSchedule' is in Cluster 2
Silhouette Score: 0.05046198517084122
Davies-Bouldin Index: 0.5768900796686108
```

Figure 4.12: K-Means Clustering Results with Silhouette Score and Davies-Bouldin Index

## 4. Glove

```
"C:\Program Files\Python312\python.exe" "C:\Users\ritik\PycharmProjects\py
Loading GloVe model...
GloVe model loaded.
Enter the path to the folder containing .docx and .doc files: C:\Users\rit

Processing Document 1...

Co-occurrence Matrix:
      human  subfield  analyzing  humans  focuses
human      0         0         0         0         0
subfield    0         0         0         0         0
analyzing   0         0         0         0         0
humans      0         0         0         0         0
focuses    0         0         0         0         0

Pairs of Words with the Highest Co-occurrence Counts:
language - nlp: 3
human - language: 2

Singleton Words (based on co-occurrence):
subfield
analyzing
humans
focuses
computers
...

All words for Document 1: ['language', 'human', 'nlp', 'subfield']

Processing Document 2...

Co-occurrence Matrix:
      human  vast  significant  comprehend  made
human      0     0         0         1         0
vast        0     0         0         0         0
significant  0     0         0         0         1
comprehend  1     0         0         0         0
made        0     0         1         0         0

Pairs of Words with the Highest Co-occurrence Counts:
processing - nlp: 2
machine - translation: 2
machine - leveraging: 2
machines - nlp: 2

Singleton Words (based on co-occurrence):
human
vast
significant
field
comprehend
...
Most occurring word in GloVe PCA: human
All words for Document 2: ['translation', 'machines', 'machine', 'leveraging', 'processing', 'nlp', 'human']

Processing Document 3...
```

```

"C:\Program Files\Python312\python.exe" "C:\Users\ritik\PycharmProjects\pythonProject5\hi_ratuki.py"
Loading GloVe model...
GloVe model loaded.
Enter the path to the folder containing .docx and .doc files: C:\Users\ritik\Downloads\ALL DOCS FOR OUTPUT

Processing Document 1...

Co-occurrence Matrix:
      human  subfield  analyzing  humans  focuses
human      0         0         0         0         0
subfield    0         0         0         0         0
analyzing   0         0         0         0         0
humans      0         0         0         0         0
focuses    0         0         0         0         0

Pairs of Words with the Highest Co-occurrence Counts:
language - nlp: 3
human - language: 2

Singleton Words (based on co-occurrence):
subfield
analyzing
humans
focuses
computers
...
Most occurring word in GloVe PCA: subfield
All words for Document 1: ['language', 'human', 'nlp', 'subfield']

```

```

Co-occurrence Matrix:
      human  sophisticated  automated  expected  efficient
human      0         0         0         0         0
sophisticated  0         0         0         0         0
automated     0         0         0         0         0
expected      0         0         0         0         0
efficient     0         0         0         0         0

Pairs of Words with the Highest Co-occurrence Counts:
processing - language: 2

Singleton Words (based on co-occurrence):
human
sophisticated
automated
expected
efficient
...
Most occurring word in GloVe PCA: human
All words for Document 3: ['language', 'processing', 'human']

```

```

Processing Document 4...

Co-occurrence Matrix:
      maintaining  wide  plenty  hydrated  water
maintaining      0    0    0         0         0
wide             0    0    0         0         0
plenty           0    0    0         1         1
hydrated         0    0    1         0         0
water           0    0    1         0         0

```

```
hydrated      0      0      1      0      0
water          0      0      1      0      0
```

Pairs of Words with the Highest Co-occurrence Counts:  
diet - balanced: 2

Singleton Words (based on co-occurrence):

```
maintaining
wide
plenty
hydrated
water
...
```

Most occurring word in GloVe PCA: maintaining

All words for Document 4: ['diet', 'balanced', 'maintaining']

Processing Document 5...

Co-occurrence Matrix:

|             | maintaining | plays | consolidate | bed | crucial |
|-------------|-------------|-------|-------------|-----|---------|
| maintaining | 0           | 0     | 0           | 0   | 1       |
| plays       | 0           | 0     | 0           | 0   | 1       |
| consolidate | 0           | 0     | 0           | 0   | 0       |
| bed         | 0           | 0     | 0           | 0   | 0       |
| crucial     | 1           | 1     | 0           | 0   | 0       |

Pairs of Words with the Highest Co-occurrence Counts:

```
sleep - habits: 2
sleep - irregular: 2
```

```
sleep - night: 2
```

Singleton Words (based on co-occurrence):

```
maintaining
plays
consolidate
bed
crucial
...
```

Most occurring word in GloVe PCA: maintaining

All words for Document 5: ['irregular', 'sleep', 'habits', 'night', 'maintaining']

Enter the number of clusters (1 to 5): 2

Document 'Doc1\_NLP' is in Cluster 0

Document 'Doc2\_AI' is in Cluster 0

Document 'Doc3\_ML' is in Cluster 0

Document 'Doc4\_Diet' is in Cluster 1

Document 'Doc5\_SleepSchedule' is in Cluster 1

Silhouette Score: 0.2259564475544221

Davies-Bouldin Index: 1.0378735757282713

Process finished with exit code 0

Figure 4.13 Output and K-Means Clustering Results with Silhouette Score and Davies-Bouldin Index

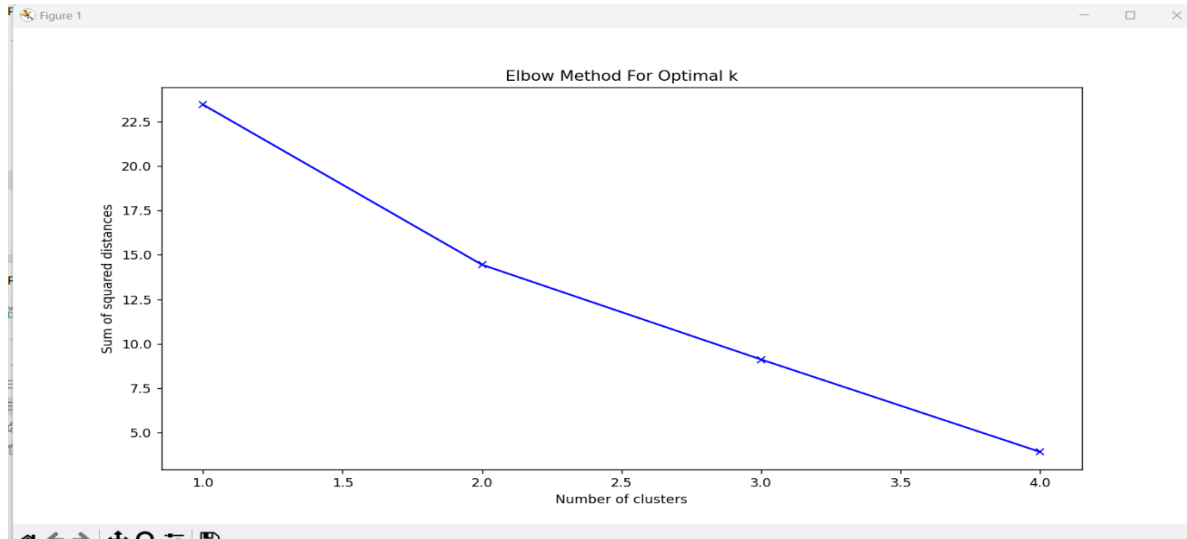


Figure 4.14 : Elbow Method for Optimal Number of Clusters

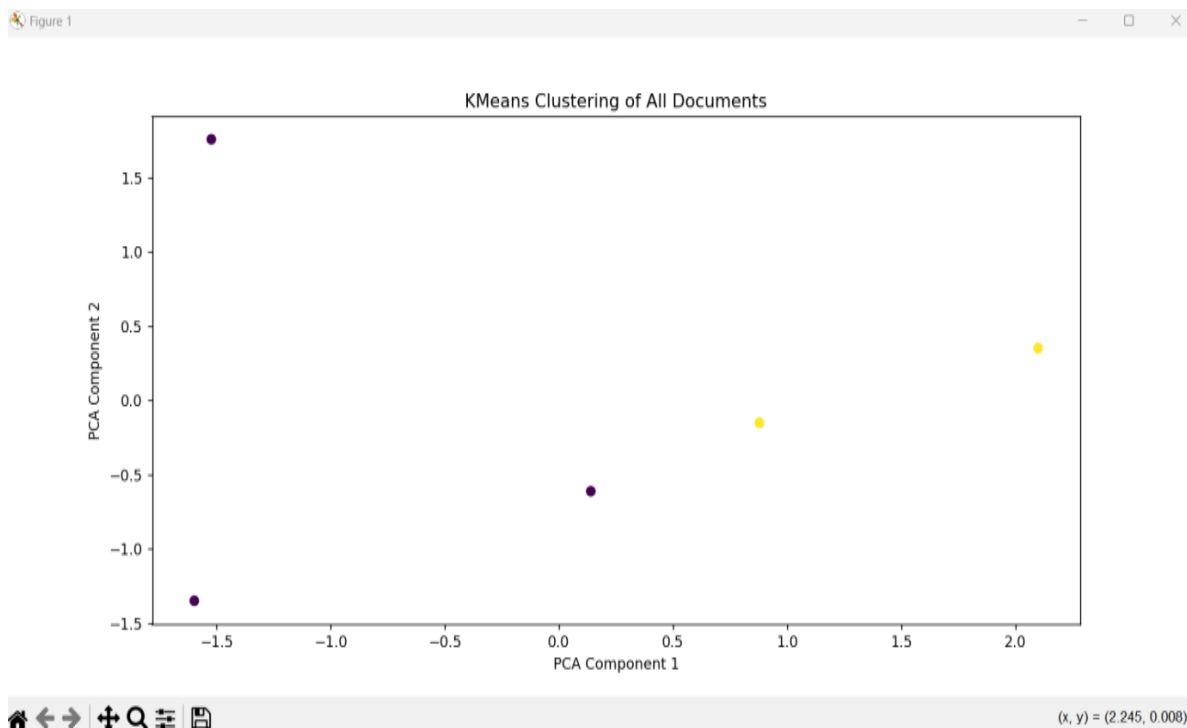


Figure 4.15 : Plot of Documents



#### 4.4 Comparative Analysis of Word Embedding Techniques

| Embedding Method | Number of Clusters | Silhouette Score | Davies-Bouldin Index | Performance     |
|------------------|--------------------|------------------|----------------------|-----------------|
| Bag of Words     | 3                  | 0.0915           | 0.5515               | <b>Poor</b>     |
| TF-IDF           | 3                  | 0.5791           | 0.6115               | <b>Best</b>     |
| Word2Vec         | 3                  | 0.0505           | 0.5769               | <b>Moderate</b> |
| GloVe            | 2                  | 0.2259           | 1.0378               | <b>Moderate</b> |

In the analysis of word embedding techniques on the dataset, various methods were evaluated to determine which offered the best representation for clustering. The Bag of Words approach performed poorly, as it relies solely on word counts and ignores semantic relationships between words, resulting in less meaningful clusters. Conversely, the TF-IDF method performed the best,. TF-IDF effectively captured the importance of terms and provided distinct clusters by highlighting significant terms and reducing the influence of common words, which resulted in better differentiation among the documents. Word2Vec showed moderate performance. Although it captured relationships between words, Word2Vec struggled to create distinct clusters for the documents, suggesting that its dense vector representations might require more sophisticated clustering algorithms for improved results. Similarly, GloVe displayed moderate performance. While GloVe captured word relationships, it also faced challenges in generating distinct clusters, indicating the need for potentially different clustering approaches to leverage its strengths fully.

# Chapter 5

## Conclusion and Future Work

### 5.1 Conclusion

This work has successfully evaluated and compared the effectiveness of various word embedding techniques, including Bag of Words, TF-IDF, Word2Vec, and GloVe, for the task of document clustering. Through extensive analysis using performance metrics such as the Silhouette Score and Davies-Bouldin Index, it was determined that TF-IDF provided the most distinct and meaningful clusters among the techniques tested. The application of co-occurrence matrices further enhanced the analysis, offering valuable insights into the relationships within the textual data. The findings indicate that while traditional methods like Bag of Words and TF-IDF still hold significant value in text analysis, advanced embedding techniques like Word2Vec and GloVe offer promising capabilities for capturing nuances in the data. Even though methods like Word2Vec and GloVe showed some performance issues due to their inability to fully exploit semantic relationships in clustering, they still provided valuable insights into word relationships. This highlights the importance of integrating semantic understanding into clustering techniques. Effective clustering not only depends on capturing semantic relationships but also on applying these insights through advanced algorithms to enhance cluster quality and separation. However, these advanced techniques may require more sophisticated clustering algorithms to fully leverage their potential. The integration of PCA with KMeans clustering provided an effective means of visualizing document similarities and validating clustering results.

## **5.2 Future work**

To further enhance the document clustering analysis, several advanced strategies can be explored. Firstly, more advanced embedding techniques like BERT, GPT, and other transformer-based models can be utilized to capture deeper relationships and contextual information in documents. Secondly, experimenting with sophisticated clustering algorithms such as DBSCAN, OPTICS, or hierarchical clustering can improve the accuracy of document grouping. Integrating deep learning-based clustering methods can also dynamically adjust to the complexity of the data, providing more precise results. Lastly, integrating the document analysis system with existing content management systems (CMS) and knowledge management platforms can enhance its utility. Developing APIs will allow seamless integration with third-party applications, broadening the system's applicability and effectiveness.

# References

- 1) Pennington, J., Socher, R., & Manning, C. D. (2014). GloVe: Global Vectors for Word Representation. In Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)
- 2) Ramos, J. (2003). Using TF-IDF to Determine Word Relevance in Document Queries. Proceedings of the First International Conference on Machine Learning.
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- 5) Arora, A. (2019). Hands-On Natural Language Processing with Python.